Interoperability Conformance Specification: colorimetricEncoding – Basic and Advanced

Contents

Foreword

This document has been prepared following the ICC Intellectual Property Policy. This policy is based on the ITU-T/ITU-R/ISO/IEC Guidelines for Implementation of the Common Patent Policy (23 April 2012), with interpretations and clarifications to make it specific to ICC. A Patent Statement and Licensing Declaration form is available.

ICC Interoperability Conformance Specifications, of which this document is an example, may be submitted to the competent ISO Technical Committee for consideration and development as an ISO document. If so, this foreword is to be replaced by the appropriate wording supplied by ISO.

Introduction

ISO 20677-1 defines specifications that provide a platform for defining extended (iccMAX) colour management profiles and systems for various colour workflow domains. It provides a platform for which domain specific specifications can be defined that make use of iccMAX extensions to the existing cross-platform profile format of ISO 15076-1. Thus there is greater flexibility for defining colour transforms and profile connection spaces to meet needs that cannot easily be met with ISO 15076-1. It is not envisioned that all colour management systems that use ISO 20677-1 will implement all the features or capabilities it specifies. Requirements specifying restrictions to iccMAX that apply to a particular workflow are defined in workflow domain specifications known as Interoperability Conformance Specifications, of which this document is one example. Additionally, for some domain specific workflows it is envisioned that workflows will connect both to profiles defined by ISO 20677-1 (iccMAX) and those defined by ISO 15076-1.

An Interoperability Conformance Specification (ICS) is approved and registered by the International Color Consortium (ICC). It defines minimum structural and operational requirements for writing and reading ICC profiles in order to address a specific problem and/or functionality that cannot readily be handled using the profile format defined by ISO 15076-1. An ICS document essentially defines restrictions to ISO 20677-1 for a specific use case.

Interoperability Conformance Specification: colorimetricEncoding – Basic and Advanced

1 Scope

All parts of this specification define scenario requirements and restrictions to profiles based on ISO 20677-1 for the purpose of defining relationships between colorimetric device encodings and a colorimetric Profile Connection Space.

The particular sub-set of the tags defined in ISO 20677-1 that are required to be present is defined, together with any optional tags that are permitted. The connections between profiles are described, and the processing elements that the CMM is required to support are identified.

This ICS defines transforms with basic limited processing element support thus enabling colorimetric encoding workflows with less implementation requirements, as well as transform that utilize advanced or extended processing element support. more complicated modeling and color management situations. The distinction between basic and advanced shall be identified by the subclass version in the profile. Some implementations may choose to only implement support for basic (subclass 1) profiles conforming to this specification.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 20677-1:20677, *Image technology colour management — Extensions to architecture, profile format, and data structure: iccMAX*

NOTE: The most recent version of the iccMAX specification is available on the ICC web site [2].

3 Terms and definitions

All terms and definitions relevant to this document are provided in ISO 20677.

3.1 observing conditions

the observer's colour matching functions and illuminant associated with the spectralViewingConditionsTag of the Profile Connection Condition (PCC) used to connect the profile to the PCS

4 Use case

4.1 Domain

Profiles and workflows conforming to this ICS shall apply to the domain of General (i.e. the domain of application is unspecified).

4.2 Intended use

The intended use of this specification is to define workflows in which colorimetric data encoding needs to be defined along with associated observer and illuminant. Custom observer and illuminant information are provided (in the spectralViewingConditionsTag) along with transforms to convert between standard (D50 for 2-degree standard observer) and custom observing conditions (via the customToStandardPccTag and the standardToCustomPccTag). Which parts of the profile that are used is dependent on which workflow scenario is used. Two major classes of workflow are enabled. The first is using transforms that provide a direct encoding of colorimetric data into and out of PCS encoding. The second is to provide Profile Connection Condition overrides that are used as part of spectral or custom colorimetric PCS operations when connecting or applying transforms of other profiles.

Note 1: Profiles conforming too this ICS can either support integer-based encoding of colorimetric values (which corresponds to how ISO 15076 based profiles encode colorimetric data) or using floating-point encoding. As defined by ISO 20677, colorimetric PCS encoding using multiProcessElement (MPE) transforms is floating-point based. Integer device-based data going into or out of a floating-point transform is mapped from 0.0 to 1.0 going into or out of the MPE transform. To work correctly, the transform needs to convert this range into the floating-point representation of the PCS. Alternatively, an MCS based transform may expect to receive floating point data coming in (not integer based) and therefore an identity transform is appropriate.

Note 2: Images embedded with profiles conforming to this specification will not work with applications that to do not support iccMAX-based (ISO 20677-1) color management.

4.3 Restrictions

ISO 20677-1 provides full details of the requirements for iccMAX profiles. This document defines a set of restrictions which apply to profiles created for the specific use cases described above. Such restrictions include the sub-set of tags from ISO 20677-1 which are permitted in profiles conforming to this document.

5 Workflow

5.1 Profile sub-classes

A supporting CMM shall support the sub-classes of the profile classes identified in Table 1 for conformance with this ICS. All profile classes are defined in ISO 20677-1.

Table 1. Sub-classes of profile classes defined by this ICS

Profile	Profile class	Sub-class Identifier	Class signature	Sub-class signature
E	colorSpace	colorimetricEncoding	‘spac’ or (73706163h)	‘pcc ’ or (70636320h)

5.2 Connection Scenarios

5.2.1 General

A ‘pcc’ colorimetricEncoding profile connects 3 input channels to a colorimetric Profile Connection Space. It can be used as either a source profile, a destination profile, or as a PCC override.

5.2.2 Profiles for ICS workflow scenarios

Table 2 provides overview information about additional profiles that are used to describe several profile connection scenarios conforming to this ICS.

Table 2. Additional profiles referenced in workflow scenarios defined by this ICS

Profile	Description
C	Profile conforming to ISO 20677-1 or ISO 15076-1 containing (a) colorimetric transform(s) with CMM configured to use a colorimetric transform type
S	Profile conforming to ISO 20677-1 containing (a) spectral reflectance transform(s) with the CMM configured to use a spectral transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.
P	Profile conforming to ISO 20677-1 containing Profile Connection Condition tags (spectralViewingConditionsTag, customToStandardPccTag, standardToCustomPccTag) Note: a profile conforming to this ICS is also an instance of a profile P.
1	Profile conforming to ISO 20677-1 or ISO 15076-1
2	Profile conforming to ISO 20677-1 utilizing Profile Connection Conditions to go into or out of the PCS

5.2.3 Workflow Scenarios

5.2.3.1 General

The following sections document scenarios that general purpose CMMs should implement to conform to this ICS. For each scenario the profile sequence is depicted along with associated CMM control parameters that select between workflow scenarios.

Note: It is recognized that specialized CMMs may choose to not implement all the scenarios in these sections due to specialized requirement needs of their supporting application(s).

5.2.3.2 Scenario 1: Connecting source profile to a colorimetricEncoding profile as destination

Figure 1 – Profile sequence for Scenario 1

In this scenario an arbitrary profile (profile 1 from Table 2) is connected to a profile conforming to this ICS (profile E from Table 1) used as a destination profile.

A CMM that supports this scenario shall be configured to process profiles in the sequence shown in Figure 1 with the associated CMM control parameters as identified in Table 3 and described by ISO 20677:2019 Annex K.

Table 3. CMM control parameters for Scenario 1

Profile	CMM Control Parameter	Value
1	Rendering Intent	Any
	Transform Type	Any
	PCC Override	Any
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	None

Device values shall first be transformed to a colorimetric PCS according to the requirements (ICS or specification) associated with profile 1.

If the observing conditions for profile E do not match those of the PCS resulting from applying profile 1 and the observing conditions for profile E are not the same as the 2-degree observer under D50 then the transform in the standardToCustomPccTag from profile E shall be applied.

The resulting colorimetry shall then be transformed by a BToAxTag from profile E (since it is using the colorimetric transform type).

5.2.3.3 Scenario 2: Connecting source profile to a colorimetricEncoding profile as destination using a PCC override

Figure 2 – Profile sequence for Scenario 2

In this scenario an arbitrary profile (profile 1 from Table 2) is connected to a profile conforming to this ICS (profile E from Table 1) used as a destination profile.

A CMM that supports this scenario shall be configured to process profiles in the sequence shown in Figure 2 with the associated CMM control parameters as identified in Table 4 and described by ISO 20677:2019 Annex K.

Table 4. CMM control parameters Scenario 2

Profile	CMM Control Parameter	Value
1	Rendering Intent	Any
	Transform Type	Any
	PCC Override	Any
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	P

Device values shall first be transformed to a colorimetric PCS according to the requirements (ICS or specification) associated with profile 1.

If the observing conditions for profile E do not match those of the PCS resulting from applying profile 1 and the observing conditions for profile E are not the same as the 2-degree observer under D50 then the transform in the standardToCustomPccTag from profile P (from Table 2) is applied (as it provides the PCC override).

The resulting colorimetry shall then be transformed by a BToAxTag from profile E (since it is using the colorimetric transform type).

5.2.3.4 Scenario 3: Connecting a colorimetricEncoding profile as source to a destination profile with a colorimetric PCS

Figure 3 – Profile sequence for Scenario 3

In this scenario a profile conforming to this ICS (profile E from Table 1) is used as a source profile with a colorimetric PCS connected to an arbitrary profile (profile C from Table 2) that is set up to use a colorimetric PCS. The transform type for profile E is “colorimetric” indicating that the transform from an AToBxTag from profile E is used to transform device values to a colorimetric PCS.

A CMM that supports this scenario shall be configured to process profiles in the sequence shown in Figure 3 with the associated CMM control parameters as identified in Table 5 and described by ISO 20677:2019 Annex K.

Table 5. CMM control parameters for Scenario 3

Profile	CMM Control Parameter	Value
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	None
C	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	Any

In this scenario either the observing conditions in profile E shall be for the standard 2-degree observer under D50 or they match those of profile C.

The resulting colorimetry from the transform in Profile E shall be transformed and applied as needed according to the requirement (ICS or specification) associated with profile C.

5.2.3.5 Scenario 4: Connecting a colorimetricEncoding profile as source using a PCC override to a destination profile with a colorimetric PCS

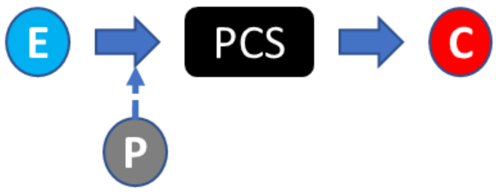

Figure 4 – Profile sequence for Scenario 4

In this scenario a profile conforming to this ICS (profile E from Table 1) is used as a source profile with a colorimetric PCS connected to an arbitrary profile (profile C from Table 2) that is set up to use a colorimetric PCS. The transform type for profile E is “colorimetric” indicating that the transform from an AToBxTag from profile E is used to transform device values to a colorimetric PCS.

A CMM that supports this scenario shall be configured to process profiles in the sequence shown in Figure 4 with the associated CMM control parameters as identified in Table 6 and described by ISO 20677:2019 Annex K.

Table 6. CMM control parameters for Scenario 4

Profile	CMM Control Parameter	Value
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	P
C	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	Any

If the observing conditions of profile E do not match those of profile C then the transform from the customToStandardPccTag in profile P (from Table 2) is applied resulting in standard colorimetry in the PCS.

The resulting colorimetry shall then transformed and applied as needed according to the requirement (ICS or specification) associated with profile C.

5.2.3.6 Scenario 5: Using a colorimetricEncoding profile as a PCC override

OR

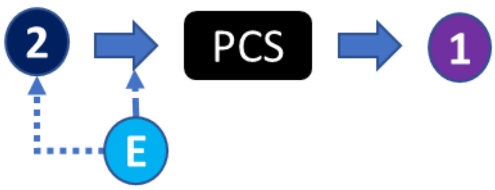

Figure 5 – Profile sequences for Scenario 5

In this scenario the profile connection conditions of a profile conforming to this specification (profile E from Table 1) are used as a PCC override of an arbitrary profile 2 (from Table 2). Profile 2 may be used as either an input or output profile when connected to some other arbitrary profile (profile 1 from Table 2).

The observing conditions from profile E may be used as part of the application of the transform (via processing elements that use a PCC observer or illuminant to perform the transform of PCS conversion/adjustment as required for profile 2. Additionally, the transform from the standardToCustomPccTag or customToStandardPccTag of profile E may be applied as required as part of the PCS conversion/adjustment for profile 2.

A CMM that supports this scenario shall be configured to process profiles in one of the sequences shown in Figure 5 with the associated CMM control parameters as identified in Table 7 and described by ISO 20677:2019 Annex K.

Table 7. CMM control parameters Scenario 5

Profile	CMM Control Parameter	Value
1	Rendering Intent:	Any
	Transform Type:	Any
	PCC Override:	Any
2	Rendering Intent:	Any
	Transform Type:	Any
	PCC Override:	E

In this scenario all other transforms of profile E shall be ignored.

5.2.3.7 Scenario 6: Converting spectral encoding to colorimetric encoding

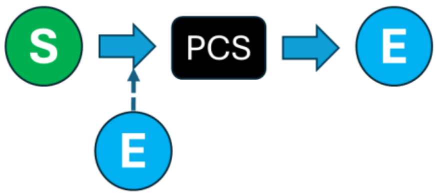

Figure 6 – Profile sequence for Scenario 6

This scenario is used to convert from device values using a spectral transform getting spectral PCS values that are converted to colorimetry using the PCC override profile (conforming to this ICS), and then output as colorimetric encoding using the destination profile (conforming to this ICS)

A CMM that supports this scenario shall be configured to process profiles in the sequence shown in Figure 6 with the associated CMM control parameters as identified in Table 8 and described by ISO 20677:2019 Annex K.

Table 8. CMM control parameters for Scenario 6

Profile	CMM Control Parameter	Value
S	Rendering Intent	Any
	Transform Type	Spectral
	PCC Override	E
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	Any

Applying this scenario involves first applying a device encoding to spectral PCS transform from a profile (profile S from Table 2) to get spectral PCS values that are then converted to a colorimetric PCS using the override observer and illuminant (observing conditions) from a PCC override profile conforming to this ICS (profile E from Table 1).

If the observing conditions of the PCC override profile E do not match those of destination profile E then the transform from the customToStandardPccTag in PCC override profile E is applied followed by transform from the standardToCustomPccTag in the destination profile E to convert the PCS to the destination observing conditions.

The resulting colorimetry shall then transformed and applied as needed using the BToAx transform in the destination profile E.

5.2.3.8 Scenario 7: Converting from one colorimetric encoding to another colorimetric encoding (with possible change in observing conditions)

Figure 7 – Profile sequence for Scenario 7

In this scenario both the source and destination profiles are instances of profiles that conform to this ICS. This scenario is used to convert from one colorimetric encoding to another. There are two functional aspects to this scenario that may apply: a change in color encoding (i.e. conversion between Lab and XYZ encoding), and a change in observing conditions (observer and or illuminant) that are associated with the colorimetry of the PCS associated with each profile.

Note: This scenario may also be considered as a more selective instance of Scenarios 1 and 2.

A CMM that supports this scenario shall be configured to process profiles in the sequence shown in Figure 1 with the associated CMM control parameters as identified in Table 9 and described by ISO 20677:2019 Annex K.

Table 9. CMM control parameters for Scenario 7

Profile	CMM Control Parameter	Value
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	None
E	Rendering Intent	Any
	Transform Type	Colorimetric
	PCC Override	None

Device values shall first be transformed a colorimetric PCS by applying the AToBxTag from the source profile E.

If the observing conditions for source profile E do not match those of the PCS resulting from applying source profile E and the observing conditions for destination profile E are not the same then the transform in the standardToCustomPccTag from source profile E shall be applied followed by the customToStandardPccTag from destination profile E shall be applied to get PCS .

The resulting colorimetry shall then be transformed by a BToAxTag from destination profile E to get the destination device color encoding.

5.3 Other scenario implementation details

Where CMM's support the application of abstract profiles they should also provide support for applying abstract class profiles as part of additional PCS processing associated with the scenarios in section 5.2.

6 Sub-class Profile Requirements

6.1 General

Requirements for iccMAX profiles conforming to this ICS document are listed here. ICC v4 profiles referred to in Section 5 above shall conform to ISO 15076-1.

6.2 Requirements

The requirements for the colorimetricEncoding sub-class of colorSpace class profiles are shown below.

The encoding of the profile header shall be as defined in ISO 20677-1, with the specific requirements shown in Table .

Table 10. Header requirements

Header field	Required content
Profile class	'spac' (73706163h)
Profile subclass	'pcc ' (70636320h)
Profile subclass major version	1 for basic encoding, 2 for advanced encoding
Profile subclass minor version	0
Profile flags	0
Device attributes	0 or 1
Data colour space	'XYZ'(58595a20h) or 'Lab'(4c616220h) matching signature of colorimetric PCS
MCS	0
Colorimetric PCS	'XYZ' (58595a20h) or 'Lab' (4c616220h) matching signature of data colour space
Spectral PCS	0
Spectral PCS range	0
Bispectral PCS	0

Full details of the encoding of the header fields in Table 8 are given in ISO 20677-1.

6.2.1 Basic Encoding Required tags

Profiles with subclass major version of 1 shall contain the tags listed in Table .

Table 11. Required tags

Tag name	Signature	Required content
profileDescriptionTag	'desc'	multiLocalizedUnicodeType structure containing invariant and localizable versions of the profile description for display
copyrightTag	'cprt'	multiLocalizedUnicodeType containing copyright information for the profile
AToB1Tag	'A2B1'	LutAToBType or multiProcessElementType using any combination of curveSetElement, matrixElement, CLUTElment, extendedCCLUTElement, tintArrayElement, calculatorElement,
BToA1Tag	'B2A1'	LutBToAType or multiProcessElementType using any combination of curveSetElement, matrixElement, clutElment, extendedClutElement, and tintArrayElement
mediaWhitePointTag	'mwpt'	XYZType containing normalized tristimulus values for media white point (usually same as illuminant)
customToStandardPccTag	'c2sp'	multiProcessElementType containing a single 3x3 matrix element Note: this tag shall not be required if the spectralViewingConditions are for D50 illuminant and standard 2-degree observer
standardToCustomPccTag	's2cp'	multiProcessElementType containing a single 3x3 matrix element Note: this tag shall not be required if the spectralViewingConditions are for D50 illuminant and standard 2-degree observer
spectralViewingConditionsTag	'svcn'	Structure defining observer, illuminant and (optionally) surround

The encoding of the tags listed in Table 9 shall be as defined in ISO 20677-1.

6.2.2 Advanced Required tags

Profiles with subclass major version of 2 shall be the same as that listed in Table 9 as defined in ISO 20677-1 with the expansion of the allowed processing elements for a multiProcessElementType in AToB1/BToA2 tags to all processing element types defined in ISO 20677-1 including the calculatorElement.

6.2.3 Example

The profiles Lab-float-D65_2deg-Part1.icc, Lab-int-D65_2deg-Part1.icc, XYZ-float-D65_2deg-Part1.icc, and Lab-int-D65_2deg-Part1.icc are encoded according to the requirements of this ICS document and is available in the IccDEV-Testing suite [3]. XML representations are also provided.

7 Conformance

A profile shall be considered to be in conformance with this ICS document if it meets the following conditions:

- The profile connects to the channels specified in section 5.
- The profile header includes the required content from Table 8.
- All required tags listed in Table 8 are present in the profile.
- The profile structure and all tags conform to ISO 20677-1.

A CMM shall be considered to be in complete conformance with this ICS if it meets the following conditions:

- The CMM can parse profiles that conform to this ICS
- The CMM supports and is capable of processing the channels specified in section 5 and any other profiles listed in Table 2 in the scenarios described in section 5.
- The CMM is able to process the tags listed in Table 9.
- When processing a profile conforming to this ICS, the CMM produces results that are a close approximation to those produced by the iccMAX Reference Implementation [3].

Bibliography

[1] ISO 20677-1:2019, *Image technology colour management — Extensions to architecture, profile format, and data structure*

[2] iccMAX http://www.color.org/iccmax/

[3] iccDEV Implementation http://www.color.org/iccmax/index.xalter#reficcmax

8 Annex A (informative) - Worked example scenarios

This annex describes the worked example scenarios provided in the iccDEV-Testing ICS package named ColorimetricEncoding. Each entry is driven by a JSON configuration file in the package's config/ directory and demonstrates one of the workflow scenarios defined in clause 5.2.3 applied to example colorimetricEncoding profiles.

Table A.1. Worked example scenarios in the iccDEV-Testing ICS package

ID	Clause 5.2.3 scenario	iccDEV tool / driver file	What is demonstrated
S1a	Scenario 1	iccApplyProfiles / ce-S1a-IccToColorEncodingD93.json	Encode an embedded-sRGB RGB image into Lab under D93 (MAT) using the colorimetricEncoding profile as the destination.
S1b	Scenario 1	iccApplyProfiles / ce-S1b-IccToColorEncodingA.json	Same as S1a but the destination encoding is Lab under Illuminant A (MAT).
S2	Scenario 2	iccApplyProfiles / ce-S2-IccToColorEncodingAWithPcc.json	Same as S1b but the destination stage carries a PCC override (Illuminant A, Abs).
S3a	Scenario 3	iccApplyProfiles / ce-S3a-ColorEncodingD93ToIcc.json	Decode the Lab(D93) output of S1a back to sRGB using the colorimetricEncoding profile as the source.
S3b	Scenario 3	iccApplyProfiles / ce-S3b-ColorEncodingAToIcc.json	Decode the Lab(A) output of S1b back to sRGB.
S3c	Scenario 3	iccApplyProfiles / ce-S3c-ColorEncodingAPccToIcc.json	Decode the PCC-overridden Lab(A) output of S2 back to sRGB.
S4	Scenario 4	iccApplyProfiles / ce-S4-ColorEncodingAWithPccToIcc.json	Same input as S3c but the source stage carries a PCC override.
S5a	Scenario 5	iccApplyProfiles / ce-S5a-IccToIccWithPcc.json	Render sRGB to a Lab(A) encoding using profile E only in its PCC-override role.
S5b	Scenario 5	iccApplyProfiles / ce-S5b-ColorIccWithPccToIcc.json	Round-trip the S5a output back to sRGB, again with profile E supplying the PCC override.
S5c	Scenario 5	iccApplyProfiles / ce-S5c-SpectralIccWithPccToIcc.json	Render a multispectral-RGB image to sRGB with profile E supplying the PCC override.
S6a	Scenario 6	iccApplyNamedCmm / ce-S6a-RefToXYZA.json	Encode chartRef.txt spectral reflectance into E-XYZ under Illuminant A via the spectral PCS profile.

S6b	Scenario 6	iccApplyNamedCmm / ce-S6b-RefToXYZD50.json	Same as S6a but destination is E-XYZ under D50.
S6c	Scenario 6	iccApplyNamedCmm / ce-S6c-RefToXYZD93.json	Same as S6a but destination is E-XYZ under D93. Output is the source for S7a and S7b.
S6d	Scenario 6	iccApplyNamedCmm / ce-S6d-RefToLabD50.json	Encode chartRef.txt as E-Lab under D50, standard 2 deg observer. Output is the source for S7c..S7f.
S6e	Scenario 6	iccApplyNamedCmm / ce-S6e-RefToLab2015.json	Encode chartRef.txt as E-Lab under D50 using the CIE 2015 2 deg observer.
S6f	Scenario 6	iccApplyNamedCmm / ce-S6f-RefToLabCat8.json	Encode chartRef.txt as E-Lab under D50 using the Asano Cat8 categorical observer.
S7a	Scenario 7	iccApplyNamedCmm / ce-S7a-XYZToLabA.json	Convert S6c output (E-XYZ encoded under D93) to E-Lab encoded under Illuminant A.
S7b	Scenario 7	iccApplyNamedCmm / ce-S7b-XYZToLabD93.json	Convert S6c output to E-Lab encoded under D93 (float Lab profile).
S7c	Scenario 7	iccApplyNamedCmm / ce-S7c-LabD50ToLabA.json	Convert S6d output (E-Lab encoded under D50) to E-Lab encoded under Illuminant A.
S7d	Scenario 7	iccApplyNamedCmm / ce-S7d-LabD50ToLabD93.json	Convert S6d output to E-Lab encoded under D93 (float Lab profile).
S7e	Scenario 7	iccApplyNamedCmm / ce-S7e-LabD50ToLab2015.json	Convert S6d output to E-Lab encoded under D50 with the CIE 2015 2 deg observer.
S7f	Scenario 7	iccApplyNamedCmm / ce-S7f-LabD50ToLabCat8.json	Convert S6d output to E-Lab encoded under D50 with the Asano Cat8 categorical observer.

NOTE: A few observations about the data produced by the worked examples in Table A.1:

Reflectance to XYZ encoding (S6a / S6b / S6c) produces a different value for the perfect reflecting diffuser under each illuminant: the encoded XYZ for the PRD is approximately the illuminant's white-point chromaticity (e.g. 1.0984 / 1.0000 / 0.3559 under Illuminant A, 0.9642 / 1.0000 / 0.8249 under D50, 0.9532 / 1.0000 / 1.4140 under D93).

Reflectance to Lab encoding (S6d / S6e / S6f) produces L*=100, a*=b*=0 for the PRD under every observer / illuminant, because the Lab encoding normalises to its destination white point by construction. Observer-induced differences only appear on chromatic patches.

Computing colorimetry from reflectance under an observer / illuminant (S7a, S7b) is not the same as converting one colorimetric encoding to another (S7c, S7d). S7a and S7b cross-convert via the source XYZ profile's chromatic-adaptation matrices, while S7c and S7d cross-convert via the destination Lab profile's customToStandardPccTag / standardToCustomPccTag, so small numerical differences are expected.

Note: In these examples (MAT) indicates that a Waypoint-based Material Adjustment Transform is used to convert between custom colorimetry and standard colorimetry (D50/2-degree standard observer). (Abs) indicates that no adjustment is made to convert between custom colorimetry and standard colorimetry.

8.1 Understanding the results

The results of S1a and S2b have images that are encoded as D93 and Illuminant A. The D50 colorimetry from the source image profile is converted by the PCC standardToCustom transforms in the colorimetricEncoding profiles.

The results of S3a and S3b should look nearly identical to the original image. This is because the PCC customToStandard transforms are being applied to get back to standard colorimetry. The result of S3c is different because a PCC override profile is provided that has an identity customToStandard transform providing no conversion of the custom colorimetry.

The results of S4 are the same as the original image because since S2 saves the image with a PCC override that doesn't convert the colorimetry, and S4 applies the same PCC override so that the colorimetry matches.

The Scenario 5 are examples of using the colorimetricEncoding profiles as PCC overrides.

Example S5a creates an image with custom colorimetry using the PCC override. Example S5b takes the image from S5a applying the PCC override inverse to get back the original colorimetry.

Example S5c demonstrates using the PCC spectralViewingConditions in the colorimetricEncoding profile to provide the observer and illuminant to spectral reflectances to get colorimetry that is then converted to standard colorimetry using the customToStandard PCC transform.

